

Paragliding Intelligence Platform

Volume 13 - Launch-Site Playbooks & Local Expert Operations

Technical Specification v2

Document field	Value
Purpose	Define the launch-site intelligence layer that converts generic forecasts into local, validated, site-specific flyability guidance.
Audience	Product, backend, GIS, weather, mobile, safety, club operations, support, legal and investor stakeholders.
Replaces	Previous high-level local-site notes and community sections.
Depends on	Volumes 01-12: PRD, SAD, Database, API, Weather, AI/ML, Mobile, Infrastructure, Security, QA, Business, Legal.
Safety posture	Decision support only. The pilot, instructor, club or event safety director remains responsible for the final decision.

Core thesis: the moat is not another weather map. The moat is trusted local intelligence per launch site.

1. Executive Summary

This volume defines the launch-site playbook and local expert operations system. The previous platform volumes describe the data, API, AI, weather and security layers. This document turns those layers into the most defensible part of the product: a site-specific operational knowledge base that knows how each launch behaves in real conditions.

Windy-style apps show weather. Paragliding-specific site apps show launch pins, forecasts, photos and webcams. The next-edge product must go further: it must know the dangerous wind directions, rotor zones, local valley-wind timing, launch closure rules, school-friendly limits, emergency access paths, landing field risks, and how much trust should be placed in each live report.

The launch-site playbook is treated as a first-class product domain, not a note field. Every launch has structured safety rules, directional envelopes, confidence thresholds, local expert validation, version history, disputes, effective dates, data provenance and machine-readable constraints consumed by the Flyability Engine, AI Briefing Engine, Mobile UI, Alerts, Club Portal and Rescue Mode.

1.1 Strategic Value

- Creates a local-data moat that is difficult for generic weather apps to copy.
- Improves safety by preventing generic model output from being interpreted without site context.
- Makes club, school and event subscriptions more valuable because they can maintain official local playbooks.
- Enables AI briefings to be grounded in verified local operating rules instead of general weather advice.
- Provides a scalable moderation system for global site intelligence without central staff manually knowing every launch.

1.2 What This Volume Specifies

- Launch-site playbook schema and lifecycle.
- Local expert, club moderator and site steward roles.
- Danger-zone geometry model for rotor, obstacles, power lines, landing hazards and access constraints.
- Wind-direction envelope and condition-rule language.
- Report validation, reputation and dispute-resolution model.
- Site-specific launch briefing generator requirements.
- Operational workflows for onboarding, editing, reviewing and retiring site rules.
- APIs, database additions and mobile UX requirements for playbooks.
- Quality gates, audit trails and safety review process.

2. Competitive Context

Current paragliding and weather products already provide broad weather maps, site databases, forecasts, webcams and flight/navigation functions. The competitive gap is not layer quantity; it is site-specific decision intelligence.

Product / ecosystem	Existing strength	Opportunity for this platform
Windy / Ventusky style weather maps	Excellent weather visualization and model layers.	Add launch-specific interpretation: what this forecast means for this exact takeoff and pilot profile.
Paragliding site-map apps	Worldwide launch/landing databases, webcams and weather	Add machine-readable playbooks, validated local expertise and

	station references.	safety-aware rule execution.
Flight computers / trackers	In-flight navigation, instruments and logging.	Integrate pre-flight site intelligence and live risk warnings from local rules.
Local club knowledge	High trust, practical, site-specific knowledge.	Digitize it into versioned rules that can be used by scoring, alerts and AI.

The platform should not attempt to replace the pilot-in-command or local club authority. It should make local safety knowledge easier to find, maintain, validate and apply consistently.

3. Domain Model Overview

The launch-site intelligence domain is built around five core objects: Launch Site, Playbook, Rule, Zone and Expert Validation. Each object is versioned and auditable.

Object	Purpose	Owned by	Consumed by
launch_site	Canonical site record with location, altitude, orientation and access metadata.	Site Service	Map, search, scoring, AI, tracking, rescue.
site_playbook	Versioned operational guide for a site.	Playbook Service	Flyability, AI briefing, mobile site detail, club portal.
condition_rule	Machine-readable rule such as max gust, unsafe wind sector, min cloudbase, no-fly season.	Playbook Service	Risk Engine, Flyability Engine, Alert Engine.
hazard_zone	PostGIS geometry representing rotor, power line, tree line, turbulence, LZ danger or access issue.	GIS Service	Map, airspace guardian, mobile warning, rescue.
expert_validation	Review/approval artifact from local expert or club officer.	Trust Service	Playbook publication, confidence score, disputes.
pilot_report	Live user observation with trust weighting.	Community Service	Reality layer, forecast correction, moderation.

3.1 Service Boundary

Launch-site playbooks are not free text stored inside a generic sites table. They require a dedicated Playbook Service because they have lifecycle, moderation, audit and safety implications.

```

playbook-service
  owns: site_playbooks, playbook_versions, condition_rules, hazard_zones,
        expert_validations, disputes, moderation_actions
  reads: launch_sites, forecasts, observations, user_roles, subscriptions
  publishes events: playbook.submitted, playbook.approved, playbook.published,
                   rule.changed, hazard_zone.changed, site_confidence.changed
  consumed by: flyability-service, ai-briefing-service, mobile-api,
              notification-service, event-ops-service, rescue-service

```

4. Launch Playbook Content Model

A playbook is the structured local operating manual for a launch site. It must be readable by humans and executable by software.

Section	Human content	Machine-readable fields
Site overview	What kind of site it is, typical season, terrain style.	site_type, difficulty, seasonality, default_skill_level.
Launch directions	Good, marginal and dangerous wind sectors.	direction_envelopes[] with speed/gust limits.
Wind behavior	Valley winds, venturi, rotor, wave/föhn notes.	microclimate_flags, risk_rules, time_of_day_modifiers.
Thermals	Known triggers, strong thermal areas, beginner cautions.	thermal_zones, expected_trigger_windows.
Landing	Primary/secondary LZ, glide constraints, obstacles.	landing_zones, lz_required_glide, lz_hazard_geometries.
Hazards	Power lines, trees, rotor, turbulence, access conflicts.	hazard_zones with severity and effective dates.
Access and rules	Parking, landowner rules, seasonal restrictions.	access_rules, closure_calendar, permit_required.
Emergency	Rescue access, local emergency contacts, extraction routes.	rescue_points, road_access, emergency_notes_visibility.

4.1 Example Playbook JSON

```
{
  "site_id": "site_9f3a2c",
  "playbook_version": 7,
  "status": "published",
  "verified_by": ["club_safety_officer", "local_expert"],
  "summary": "Mountain ridge launch with strong valley-wind sensitivity.",
  "default_recommendation_mode": "conservative",
  "direction_envelopes": [
    {
      "sector_deg": [300, 030],
      "label": "preferred",
      "min_wind_kmh": 6,
      "max_wind_kmh_beginner": 16,
      "max_wind_kmh_intermediate": 22,
      "max_gust_kmh": 28,
      "max_gust_spread_kmh": 10
    },
    {
      "sector_deg": [240, 300],
      "label": "marginal_crosswind",
      "requires_expert_review": true
    },
    {
      "sector_deg": [180, 240],
      "label": "danger_rotor_sector",
      "hard_blocker": true
    }
  ],
  "rules": [
    {
      "rule_id": "gust_spread_blocker",
      "if": "gust_spread_kmh >= 14",
      "then": "NO_GO",
      "severity": "hard"
    }
  ],
}
```

```

{
  "rule_id": "low_cloudbase_warning",
  "if": "cloudbase_agl_m < 300",
  "then": "NO_GO",
  "severity": "hard"
}
]
}

```

5. Condition Rule Language

Rules must be strict enough to run safely in the Flyability Engine but flexible enough for clubs to express local site behavior. The rule language should start as a constrained JSON DSL, not arbitrary code.

Rule type	Example	Execution behavior
hard_blocker	gust_spread_kmh >= 14	Overrides all positive scores and returns NO-GO with explanation.
score_penalty	model_disagreement_index > 0.55	Reduces confidence or Flyability score.
skill_gate	max_wind_kmh depends on pilot_level	Applies different thresholds for beginner, intermediate, advanced, tandem, school.
time_window	valley_wind_after_local_time >= 16:00	Adds risk during time ranges.
seasonal_rule	no_fly dates or nesting period	Blocks or warns during configured date ranges.
live_confirmation_required	forecast confidence low and no station report	Returns MAYBE until recent verified observation exists.

5.1 Rule DSL Constraints

- Rules can only reference approved variables from the canonical weather and site-feature dictionary.
- No arbitrary Python/JavaScript execution is allowed.
- All rule updates require validation against test scenarios before publication.
- Every hard blocker must include a human-readable explanation and citation/provenance.
- Rules have effective_from and effective_to dates to handle temporary closures or seasonal landowner restrictions.
- Rules must declare impacted pilot profiles: all, beginner, school, tandem, XC, competition or rescue.

5.2 Rule Evaluation Order

1. Load published playbook for site and requested time window.
2. Load user pilot profile and subscription entitlement context.
3. Load forecast cube, model disagreement, live stations and recent reports.
4. Evaluate hard blockers first.
5. Evaluate skill gates and school/tandem restrictions.
6. Apply scoring penalties and confidence modifiers.
7. Generate structured explanation trace for AI and UI.
8. Store decision trace for audit and QA sampling.

```

decision_trace = {
  "site_id": "site_9f3a2c",

```

```

"valid_time": "2026-06-05T13:00:00Z",
"base_score": 78,
"hard_blockers": [],
"penalties": [
  {"rule_id": "model_disagreement", "delta": -12},
  {"rule_id": "valley_wind_time", "delta": -8}
],
"final_status": "MAYBE",
"confidence": 0.61,
"explanation": "Forecast is promising but model disagreement and late-day valley wind risk require live confirmation."
}

```

6. Hazard Zone Geometry Model

Hazards should be represented as geometry, not text. This allows the mobile map, in-flight warnings, rescue mode and route planner to use the same authoritative site intelligence.

Hazard type	Geometry	Severity	Example product use
rotor_zone	Polygon or multipolygon	warning / dangerous / hard	Show map overlay and penalize launch when wind sector activates rotor.
power_line	LineString with buffer	dangerous	Warn route planner and landing approach.
tree_line	LineString / polygon	warning	Landing field caution.
turbulence_band	Polygon	warning / dangerous	In-flight and XC route warning.
landing_obstacle	Point/line/polygon	warning / dangerous	LZ briefing and glide-to-landing checks.
access_closure	Polygon/route	informational / hard	Access and parking warnings.
emergency_access	Point/LineString	informational	Rescue mode and event safety planning.

6.1 PostGIS Data Model Extension

```

CREATE TABLE site_hazard_zones (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  site_id UUID NOT NULL REFERENCES launch_sites(id),
  playbook_version_id UUID NOT NULL,
  hazard_type TEXT NOT NULL CHECK (hazard_type IN (
    'rotor_zone', 'power_line', 'tree_line', 'turbulence_band',
    'landing_obstacle', 'access_closure', 'emergency_access', 'other'
  )),
  severity TEXT NOT NULL CHECK (severity IN ('info', 'warning', 'danger', 'hard_blocker')),
  name TEXT NOT NULL,
  description TEXT,
  geometry GEOMETRY(GEOMETRY, 4326) NOT NULL,
  activation_rule_id UUID,
  source_type TEXT NOT NULL,
  source_reference TEXT,
  verified_status TEXT NOT NULL DEFAULT 'draft',
  effective_from TIMESTAMPTZ NOT NULL DEFAULT now(),
  effective_to TIMESTAMPTZ,
  created_by UUID NOT NULL,
  created_at TIMESTAMPTZ NOT NULL DEFAULT now(),
  updated_at TIMESTAMPTZ NOT NULL DEFAULT now()
)

```

```
);
CREATE INDEX idx_site_hazard_zones_site ON site_hazard_zones(site_id);
CREATE INDEX idx_site_hazard_zones_geom ON site_hazard_zones USING GIST(geometry);
```

7. Local Expert Trust Model

Local site knowledge is valuable only if the system knows who supplied it and how trustworthy it is. The platform should separate ordinary user reports from verified local expert statements and official club rules.

Role	Can do	Cannot do	Trust weight
ordinary_pilot	Submit reports, photos, comments, suggested edits.	Publish playbooks or hard blockers.	Low to medium depending on reputation.
verified_local_pilot	Validate weather behavior, comment on rules, verify reports.	Override club official rules.	Medium to high.
site_steward	Maintain playbook drafts, hazard geometry and access notes.	Publish high-risk changes alone.	High for one or more assigned sites.
club_safety_officer	Approve/publish club-owned playbooks and temporary closures.	Edit sites outside authority without review.	Very high.
school_admin	Define school-specific student thresholds and training windows.	Publish public rules unless also club-authorized.	High inside school context.
platform_safety_admin	Intervene, lock rules, approve emergency corrections.	Act as local authority without provenance.	System authority with audit requirement.

7.1 Expert Verification Requirements

- A local expert must be linked to a real user account with verified email and optional phone verification.
- Verification can be granted by a club admin, school admin, event organizer or platform safety admin.
- Expert authority is scoped by site, region, club, school or event; never global by default.
- Every expert action is logged with actor, role, source IP/device, timestamp and affected playbook version.
- Experts can be suspended and their previous changes can be marked for review without deleting audit history.

7.2 Reputation Scoring

```
expert_trust_score = clamp(
  0.35 * verified_role_weight +
  0.20 * report_accuracy_score +
  0.15 * peer_confirmations +
  0.10 * club_affiliation_weight +
  0.10 * tenure_weight +
  0.10 * dispute_resolution_score -
  false_report_penalty,
  0.0, 1.0
)
```

Reputation must never make a dangerous report automatically safe. Trust can increase confidence in observations, but hard safety blockers and official rules remain dominant.

8. Playbook Lifecycle

Playbooks should move through formal states because they influence safety decisions.

State	Meaning	Allowed transitions
draft	New or edited playbook not visible to normal users.	submitted, archived.
submitted	Ready for review by local expert or club officer.	needs_changes, approved, rejected.
needs_changes	Reviewer requested corrections.	draft, submitted.
approved	Reviewed but not yet public.	published, draft.
published	Current production playbook used by scoring and AI.	superseded, emergency_locked.
emergency_locked	Temporarily locked because of safety dispute or incident.	published, superseded.
superseded	Replaced by a newer published version.	archived.
archived	Historical only.	none.

8.1 Publication Gates

9. Schema validation succeeds.
10. All hard blockers have explanations.
11. All direction envelopes use valid compass sectors and non-overlapping priority rules.
12. Hazard geometries pass topology validation.
13. No unresolved high-severity disputes exist.
14. At least one approved reviewer signs off for public MVP sites; two reviewers for school/event/rescue official use.
15. Regression test pack is executed against representative weather scenarios.
16. AI briefing preview is generated and checked for safe wording.

9. Local Reality Layer Operations

The reality layer combines live stations, webcams, pilot reports and expert validation. It should show the difference between forecast and reality rather than hiding uncertainty.

Input	Freshness target	Trust notes	Use
weather_station	1-10 minutes	High if calibrated and geolocated near launch.	Nowcast, alerting, forecast validation.
pilot_report	0-60 minutes	Weighted by reputation and consistency.	Reality badge, live flyability modifier.
webcam_cv	5-15 minutes	Medium; computer-vision confidence displayed.	Cloud/visibility/windsock inference.
club_notice	Immediate	Very high if official.	Temporary closures and safety warnings.
event_safety_notice	Immediate during event	Very high in event context.	Competition decisions and tracking alerts.

9.1 Report Validation Pipeline

```
pilot submits report
-> schema validation
-> spam and abuse filter
-> location proximity check
-> time freshness check
-> compare with station/model/webcam
-> assign trust score
-> publish as unverified or verified
-> update live reality layer
-> feed forecast validation after delay
```

9.2 Report Object

```
{
  "report_id": "obs_123",
  "site_id": "site_9f3a2c",
  "reported_at": "2026-06-05T12:41:00Z",
  "observer_role": "verified_local_pilot",
  "location_context": "launch",
  "wind_speed_kmh": 18,
  "gust_kmh": 25,
  "wind_direction_deg": 315,
  "cloudbase_agl_m": 900,
  "thermal_strength_label": "moderate",
  "traffic_light": "flyable_for_intermediate",
  "confidence": 0.82,
  "visibility": "public_aggregated",
  "media_ids": ["img_91a"],
  "moderation_status": "published"
}
```

10. Site-Specific AI Briefing Grounding

AI briefings must not invent local site rules. The AI Briefing Service should receive a compiled playbook context with provenance, then generate a human-friendly answer from that context only.

Context block	Source	LLM permission
Published playbook summary	playbook-service	Can summarize.
Condition rules and blockers	playbook-service/risk-engine	Can explain but cannot override.
Forecast and reality layer	weather/risk services	Can summarize values and uncertainty.
Pilot profile	profile-service	Can personalize thresholds.
Unverified report	community-service	May mention as unverified only.
Expired draft rule	playbook-service	Must not present as current.
Unknown site behavior	none	Must say unknown/no verified local playbook.

10.1 Briefing Policy

- The model must always expose uncertainty and confidence when model disagreement or missing live data exists.
- The model must not say safe; preferred terms are GO, MAYBE, NO-GO and decision support wording.
- For beginner and school users, conservative defaults apply when playbook confidence is incomplete.
- A GO result is blocked if any hard blocker exists, even if the weather looks good generally.

- If the playbook is missing for a launch, the AI must say that local site rules are not verified.

SYSTEM POLICY EXCERPT

You are a paragliding weather briefing assistant. You must only use the provided forecast, playbook, rules, live observations and pilot profile. If no verified playbook exists, state that local site intelligence is incomplete. Do not invent rotor zones, launch rules, legal access rules, emergency contacts or weather observations. Always include the main risk, confidence and best time window if available.

11. APIs

The APIs below extend Volume 04 and should be added to the OpenAPI specification.

Endpoint	Method	Purpose	Tier / role
/sites/{id}/playbook	GET	Read published playbook.	Free basic; Pro gets full rule explanations.
/sites/{id}/playbook/draft	POST	Create or edit draft playbook.	site_steward, club_admin.
/playbooks/{id}/submit	POST	Submit for review.	site_steward.
/playbooks/{id}/approve	POST	Approve playbook version.	club_safety_officer, platform_safety_admin.
/sites/{id}/hazards	GET	Fetch hazard geometries.	Free basic; Pro map overlays.
/sites/{id}/reports	POST	Submit pilot observation.	authenticated user.
/sites/{id}/local-briefing	GET	Compiled local playbook + forecast briefing.	Pro/Pro+; clubs/schools.
/sites/{id}/disputes	POST	Open safety dispute.	verified users; club roles.
/expert-verifications	POST	Grant/revoke local expert role.	club_admin/ platform_admin.

11.1 Local Briefing Response Example

```
{
  "site_id": "site_9f3a2c",
  "playbook_version": 7,
  "briefing_time": "2026-06-05T08:00:00Z",
  "status": "MAYBE",
  "confidence": 0.63,
  "best_window_local": "12:30-15:30",
  "primary_risks": [
    "Late-day valley wind likely after 16:00",
    "Model disagreement on wind speed is high",
    "No verified launch report in the last 60 minutes"
  ],
  "pilot_specific_note": "For beginner pilots this site is NO-GO today because gust spread exceeds the beginner threshold.",
  "required_live_confirmation": true,
  "provenance": {
    "playbook": "published/local_expert_verified",
    "forecast": "multi_model",
    "live_reports": "unverified_recent"
  }
}
```

12. Database Additions

These tables extend Volume 03. They should be owned by Playbook Service and Trust Service.

```
CREATE TABLE site_playbooks (  
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  site_id UUID NOT NULL REFERENCES launch_sites(id),  
  current_published_version_id UUID,  
  owner_type TEXT NOT NULL CHECK (owner_type IN ('community','club','school','event','platform')),  
  owner_id UUID,  
  confidence_score NUMERIC(4,3) NOT NULL DEFAULT 0.000,  
  created_at TIMESTAMPTZ NOT NULL DEFAULT now(),  
  updated_at TIMESTAMPTZ NOT NULL DEFAULT now(),  
  UNIQUE(site_id)  
);  
  
CREATE TABLE playbook_versions (  
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  playbook_id UUID NOT NULL REFERENCES site_playbooks(id),  
  version_number INT NOT NULL,  
  status TEXT NOT NULL,  
  summary TEXT,  
  body_markdown TEXT,  
  rules_json JSONB NOT NULL DEFAULT '[]'::jsonb,  
  source_register JSONB NOT NULL DEFAULT '[]'::jsonb,  
  created_by UUID NOT NULL,  
  submitted_at TIMESTAMPTZ,  
  approved_at TIMESTAMPTZ,  
  published_at TIMESTAMPTZ,  
  superseded_at TIMESTAMPTZ,  
  UNIQUE(playbook_id, version_number)  
);  
  
CREATE TABLE playbook_reviews (  
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  playbook_version_id UUID NOT NULL REFERENCES playbook_versions(id),  
  reviewer_id UUID NOT NULL,  
  reviewer_role TEXT NOT NULL,  
  decision TEXT NOT NULL CHECK (decision IN ('approve','reject','needs_changes')),  
  safety_notes TEXT,  
  created_at TIMESTAMPTZ NOT NULL DEFAULT now()  
);
```

13. Mobile UX Requirements

The mobile app must show site intelligence without overwhelming the pilot. The goal is fast comprehension under pre-flight pressure.

Screen area	Required UX	Safety detail
Site header	GO/MAYBE/NO-GO, confidence, last updated.	Never show GO without confidence and main risk.
Wind envelope card	Allowed directions and speed/gust limits by pilot level.	Highlight current forecast position inside/outside envelope.
Hazard map	Rotor, obstacles, LZ hazards, emergency access.	Hazards remain visible offline for saved sites.
Local notes	Short verified notes first; expandable detail.	Draft/unverified notes clearly labeled.
Reality panel	Station, webcam, pilot reports, model-vs-reality.	Freshness and trust indicators required.
Dispute/report button	Report wrong info, unsafe	High-severity reports can trigger

	condition or access issue.	admin review.
--	----------------------------	---------------

13.1 Offline Behavior

- Saved favorite sites cache the latest published playbook, hazard geometries and emergency access points.
- Offline playbooks must show last-sync age prominently.
- Offline mode cannot show current GO status unless forecast and live data are fresh enough according to configured TTL.
- If data is stale, the app displays PLAYBOOK ONLY - WEATHER DATA STALE rather than a current recommendation.

14. Club, School and Event Operations

Local organizations are the highest-trust operators. The product should give them tools that make official local knowledge easier to maintain than PDF documents or social-media posts.

Operator	Features	Business value
Club	Official playbook ownership, member alerts, closure notices, local expert roles.	Club subscription and trusted regional expansion.
School	Student-specific thresholds, instructor notes, training site calendars.	School tier and reduced operational risk.
Event organizer	Task-site playbooks, temporary hazard zones, event safety notices.	Event-tier revenue and live competition utility.
Rescue authority	Emergency access points, wind-reconstruction context, landing-zone notes.	Enterprise/rescue partnership.

14.1 Temporary Safety Notices

Temporary notices are not normal comments. They are structured operational controls with severity, effective dates and target audiences.

```
{
  "notice_type": "temporary_closure",
  "site_id": "site_9f3a2c",
  "severity": "hard_blocker",
  "effective_from": "2026-06-07T00:00:00+02:00",
  "effective_to": "2026-06-10T23:59:00+02:00",
  "reason": "Landowner access restriction",
  "issued_by_role": "club_safety_officer",
  "visible_to": ["all_users"],
  "requires_acknowledgement": false
}
```

15. Onboarding Pipeline for New Launch Sites

17. Create candidate site from trusted import, user submission or club upload.
18. Run geospatial validation: coordinate sanity, altitude from DEM, duplicate detection and access route proximity.

19. Collect minimum launch metadata: orientation, altitude, landing zone and safety note.
20. Mark as unverified community site until local validation is completed.
21. Invite local experts or relevant club to claim/verify the site.
22. Create first playbook draft using structured wizard.
23. Run rule and geometry validation.
24. Publish with confidence level: low, medium or verified.
25. Schedule periodic review, at least annually for popular sites and after incidents/disputes.

15.1 Minimum Viable Playbook

Field	Required for MVP?	Reason
launch coordinates	Yes	Map and weather extraction.
launch orientation	Yes	Direction compatibility.
safe wind range	Yes	Basic flyability.
max gust or gust spread	Yes	Safety blocker.
primary landing area	Yes	Basic flight feasibility.
main hazards	Yes	Safety briefing.
local verification status	Yes	Trust and UI labeling.
full XC playbook	No	V1/V2 enhancement.
computer-vision webcam validation	No	Advanced feature.

16. Data Sources and Licensing Controls

The system should record provenance for every imported or user-contributed site fact. External data should be imported only if license terms permit commercial use, transformation and redistribution in the planned product context.

Source class	Potential source	Control
Aeronautical / airspace	OpenAIP API and exports.	Store source attribution, update timestamp and license reference.
Map / POI / routes	OpenStreetMap via Overpass or extracts.	Respect ODbL obligations and maintain attribution/provenance.
Existing paragliding site DB	Community or commercial site databases.	Import only with explicit permission or compatible license.
Club data	Official local club uploads.	Contractual authority and role-based editing.
Pilot reports/photos	User-generated content.	Terms of contribution, moderation and deletion workflow.

Official references for design assumptions include OpenAIP documentation for aeronautical data APIs, OpenStreetMap/Overpass documentation for geodata querying, Paragliding Map public descriptions showing the existing site/weather/webcam benchmark, and FAI/CIVL safety material emphasizing risk management in hang gliding and paragliding.

17. QA and Safety Review

Site playbooks are safety-sensitive content. They require specialized test cases in addition to normal API and UI tests.

Test area	Example test	Acceptance criterion
Rule engine	Danger wind sector returns NO-GO even with high thermal/XC score.	Hard blocker always wins.
Skill personalization	Beginner sees NO-GO while advanced sees MAYBE under same conditions.	Thresholds match playbook.
Hazard geometry	Rotor polygon appears when wind sector activates.	Map and decision trace match rule.
Draft isolation	Draft playbook never affects public scoring.	Only published version used.
AI grounding	AI refuses to invent local rule when playbook missing.	Response states missing verified local playbook.
Dispute lock	Emergency lock hides GO status.	User sees safety review notice.
Offline stale data	Cached playbook older than TTL cannot show current GO.	Displays stale-data warning.

18. MVP Implementation Slice

The MVP should not try to build every possible local-knowledge feature. It should implement the smallest safe slice that makes site-specific flyability real.

MVP component	Scope
Structured playbook	Orientation, wind limits, gust limits, hazards, landing notes, access notes, verification status.
Rule execution	Hard blockers, score penalties, skill gates and stale-data blockers.
Expert roles	Platform admin, site steward, verified local pilot, club safety officer.
Moderation	Draft -> submitted -> approved -> published lifecycle.
Mobile UI	Playbook card, wind envelope, local hazards, freshness/trust badges.
AI grounding	Briefing uses only published playbook and forecast decision trace.
Audit	Every rule and geometry change logged.

18.1 V1 / V2 Expansion

- V1: club playbook ownership, local expert reputation scoring, report validation, dispute workflows.
- V1: offline saved-site playbooks and hazard geometry cache.
- V2: webcam computer vision for windsock/cloud/visibility hints.
- V2: automatic playbook recommendations from forecast-error and incident history, requiring human approval.
- V2: launch-specific digital twin dashboards and annual safety review reports.
- V3: regional expert networks, multilingual playbooks and federation/club integrations.

19. Engineering Backlog

Epic	User story / technical task	Priority
PB-001	Implement playbook-service with draft/version/publication lifecycle.	P0
PB-002	Implement condition-rule JSON DSL and evaluator.	P0
PB-003	Add site_hazard_zones table with PostGIS geometry validation.	P0
PB-004	Add expert roles and scoped authority.	P0
PB-005	Expose GET /sites/{id}/playbook and /local-briefing.	P0
PB-006	Integrate playbook hard blockers into flyability-service.	P0
PB-007	Mobile playbook card and wind envelope UI.	P0
PB-008	Pilot report validation and reputation scoring.	P1
PB-009	Club admin playbook editor.	P1
PB-010	Dispute workflow and emergency lock.	P1
PB-011	Offline site-playbook cache.	P1
PB-012	Local expert dashboard and annual review workflow.	P2
PB-013	Webcam CV integration into local reality layer.	P2

20. Acceptance Criteria

- A launch site can have a published playbook with structured rules and hazard geometries.
- Flyability decisions consume hard blockers from the playbook and expose the rule trace.
- AI briefing is grounded in the published playbook and refuses to invent missing local facts.
- Mobile users can see current recommendation, wind envelope, main local risks and data freshness.
- Club or platform roles can edit, submit, approve and publish playbooks with audit history.
- Hazard zones render on the map and can be used by route/airspace/in-flight warnings.
- Unverified reports do not silently override verified local rules.
- Offline playbook mode never presents stale weather as a current GO decision.

21. References

- OpenAIP API Documentation - <https://docs.openaip.net/>
- OpenAIP Documentation - <https://www.openaip.net/docs>
- OpenStreetMap Overpass API - https://wiki.openstreetmap.org/wiki/Overpass_API
- Overpass API service - <https://overpass-api.de/>
- Paragliding Map public product description - <https://www.paraglidingmap.com/>
- Paragliding Map App Store description - <https://apps.apple.com/us/app/paragliding-map/id449161459>
- FAI Safety Handbook / CIVL safety material - <https://www.fai.org/page/safety-handbook>

- FAI CIVL Sporting Code Section 7 guidelines and templates -
https://www.fai.org/sites/default/files/civl/documents/sporting_code_s7_i_-_guidelines_and_templates_2024.pdf